



Frequency and voltage-dependent dielectric, electric modulus, and AC conductivity analysis of Au/NAMA/n-Si MIS diode structures

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Received: 8 April 2026 / Accepted: 5 June 2026
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Abstract

The dielectric response and electrical transport characteristics of an Au/NAMA/n-Si/In MIS-type diode structure were investigated in the frequency range of 10 kHz–1 MHz and voltage range of -2 V to $+2$ V. The NAMA organic interlayer was deposited onto the n-Si substrate by the spin-coating technique. Frequency-dependent capacitance and conductance measurements were used to determine the dielectric permittivity (ϵ' , ϵ''), loss tangent ($\tan\delta$), electric modulus (M' , M''), and AC conductivity (σ_{ac}) parameters of the fabricated structure. The obtained results reveal a strong frequency dispersion behavior associated with interfacial polarization and interface-related charge effects. The ϵ' values decrease with increasing frequency, while ϵ'' and $\tan\delta$ exhibit relatively high values in the low-frequency region due to dielectric loss mechanisms. Electric modulus analysis indicates the presence of relaxation processes and non-Debye type behavior within the structure. In addition, AC conductivity increases with frequency and reaches approximately $6 \cdot 10^{-5}$ S/cm at 1 MHz. The results demonstrate that the NAMA interlayer significantly modifies the dielectric and electrical characteristics of the MIS structure and may provide potential advantages for frequency-dependent semiconductor device applications.

Keywords MIS diode · NAMA · Dielectric properties · Electric modulus · AC conductivity

1 Introduction

Organic semiconductor materials are attracting significant interest in optoelectronic technologies and research due to their suitability for diode production techniques and their ability to tunable electronic properties [1–4]. Today, organic thin films prepared using the spin coating method are used in many application areas such as diodes, sensors, and field-effect transistors [5–7]. In recent years, there has been a significant increase in studies focused on understanding the frequency-dependent electrical response of metal–insulator–semiconductor (MIS) structures. These characteristics play a crucial role in determining device performance under real operating conditions. In particular, parameters such as dielectric permittivity, loss tangent, and AC conductivity

provide detailed insight into charge transport mechanisms, interface states, and polarization processes. These parameters are of critical importance for high-frequency and optoelectronic applications.

One of the fundamental factors determining the performance of devices used in these application areas is the physical and electrical mechanisms occurring at the metal–semiconductor interface. MIS structures enable the optimisation of interface parameters by adding a thin intermediate layer to the metal–semiconductor contact.

With MIS-type structures, investigating the effect of the interface layer on charge transport mechanisms is quite popular [8–10]. The intermediate layer can directly affect fundamental electrical parameters such as barrier height, series resistance, and interface state density [11–13]. Organic intermediate layers, in particular, play a major role in reducing interface defects and improving electrical properties.

It is quite common to examine dielectric properties as a function of frequency and voltage to obtain detailed information about interface states and polarisation mechanisms in semiconductors [14–17]. Furthermore, the real part of complex dielectric constant (ϵ'), loss tangent ($\tan\delta$), electric modulus (M'), and alternating current conductivity (σ_{ac})

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parameters are frequently used to determine the physical mechanisms of diodes [18–22].

9-[(5-nitro pyridine-2-aminoethyl) iminomethyl]-anthracene (NAMA) is an organic material that has been the subject of a wide range of studies. Akin and colleagues investigated the optical properties of the NAMA organic material and its use in diode applications, calculating the ideality factor and barrier height values of the Au/NAMA/n-Si/In Schottky diode they produced as 2.61 and 0.724 eV, respectively [23]. Varol and colleagues investigated the electroluminescence behaviour of NAMA as a polycyclic aromatic hydrocarbon (PAH)-based material in organic light-emitting diodes (OLEDs) and found that NAMA exhibited blue/near-UV emission with a high-intensity signal at 410 nm [24]. Similarly, Tuğluoğlu and colleagues investigated the temperature-dependent barrier properties of the Au/NAMA/n-Si Schottky diode in the low-temperature range [25], while Eymur and colleagues examined the variation in diode parameters under different light intensities in Schottky diodes with NAMA interface layers, considering the potential applications of these structures in optoelectronic devices [26].

In addition to NAMA-based structures, various organic interlayer-based MIS and Schottky structures such as perylene derivatives, BODIPY-based compounds, and polymer-based thin films have been widely investigated [1, 7, 11, 12]. These studies demonstrate that the interlayer material has a significant influence on dielectric relaxation behavior, interface state density, and charge transport mechanisms. However, despite these extensive efforts, a comprehensive understanding of the combined effects of frequency and applied voltage on dielectric parameters and electric modulus behavior in NAMA-based MIS structures remains limited.

Numerous studies on dielectric and electric modulus properties using different organic semiconductors are available in the literature. However, studies that jointly examine the frequency- and voltage-dependent dielectric parameters, electric modulus behaviour, and AC conductivity mechanisms in MIS structures with NAMA interlayers are limited. Therefore, investigating the effect of the NAMA layer on the interface properties of such structures is of importance. In this study, an Au/NAMA/n-Si/In MIS-type diode structure was fabricated, and the dielectric properties of the structure were examined in detail in the frequency range of 10 kHz–1 MHz and the voltage range of –2 V to +2 V.

This study provides a comprehensive investigation of the frequency- and voltage-dependent dielectric properties,

electric modulus behavior, and AC conductivity mechanisms of the Au/NAMA/n-Si MIS-type diode structure. Unlike many previous studies that primarily focus on individual electrical parameters, this work systematically correlates dielectric parameters with modulus formalism and conductivity analysis to reveal the underlying charge transport and relaxation mechanisms. The key novelty of this study lies in the combined evaluation of these parameters within a unified framework, enabling a more detailed interpretation of interface-related phenomena and polarization processes in NAMA-based MIS structures.

2 Experimental

An n-Si wafer with (100) orientation, approximately 300 μm thick, and 20 $\Omega\text{ cm}$ resistivity was utilized to create an Au/NAMA/n-Si MIS type diode. Prior to the formation of contacts, the Si substrate was cleaned using the RCA chemical cleaning procedure [28]. The ohmic contact was formed by evaporating indium metal on the back of the Si substrate, which was subsequently annealed for three minutes at 380 $^{\circ}\text{C}$ in a N_2 atmosphere. The synthesis of the organic semiconductor NAMA molecule was carried out according to the procedures outlined in the literature [29], and the structural characterisation was verified by ^1H and ^{13}C spectroscopy. After the development of ohmic contact, NAMA material was dissolved in chloroform solvent (10 mg of organic material dissolved in 1 ml of chloroform solvent). A NAMA thin layer was then produced using a spin coating device on the polished front surface of silicon crystals and allowed to dry. The first thirty seconds of the NAMA thin film coating procedure were carried out at 500 rpm, and the next thirty seconds were carried out at 1200 rpm. Rectifier contact generation was achieved by depositing a 1 mm radius, 150 nm Au metal layer by thermal evaporation on a NAMA organic thin film surface. The thickness of the NAMA thin film was calculated as about 330 nm from capacitance-voltage ($C-V$) measurements at high frequencies. The diode area was calculated as $3.14 \times 10^{-6}\text{ m}^2$. The pressure used for each evaporation process was around 10^{-6} Torr. The fabricated Au/NAMA/n-Si/In diode's capacitance-voltage ($C-V$) and conductance-voltage ($G-V$) measurements of the fabricated Au/NAMA/n-Si/In diode were carried out in the dark using a Sourcetronic ST2826 LCR meter in the frequency range of 10 kHz–1 MHz and voltage range of –2 V to +2 V at room temperature. The schematic representation of the fabricated device is shown in Fig. 1.

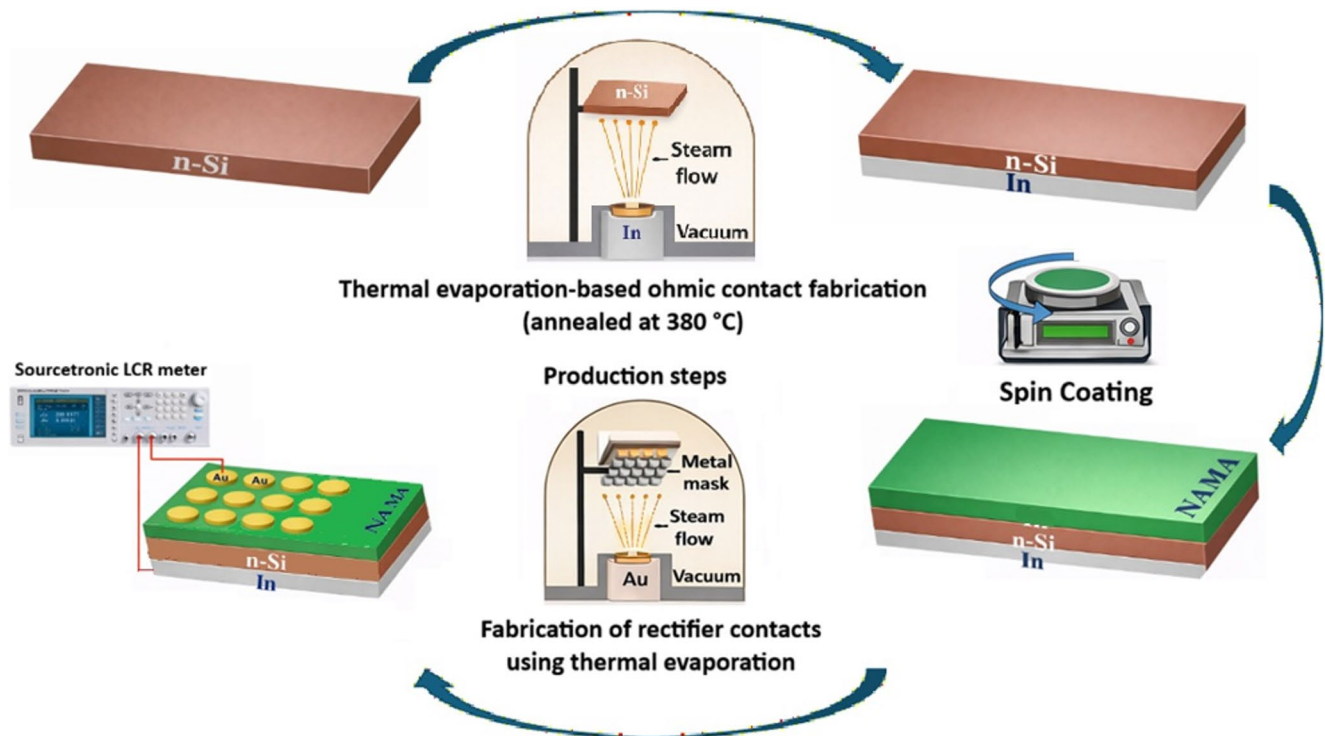


Fig. 1 Schematic diagram of Au/NAMA/n-Si/In Schottky diode fabrication procedure

3 Result and discussion

3.1 Capacitance–voltage characteristics and interface effects

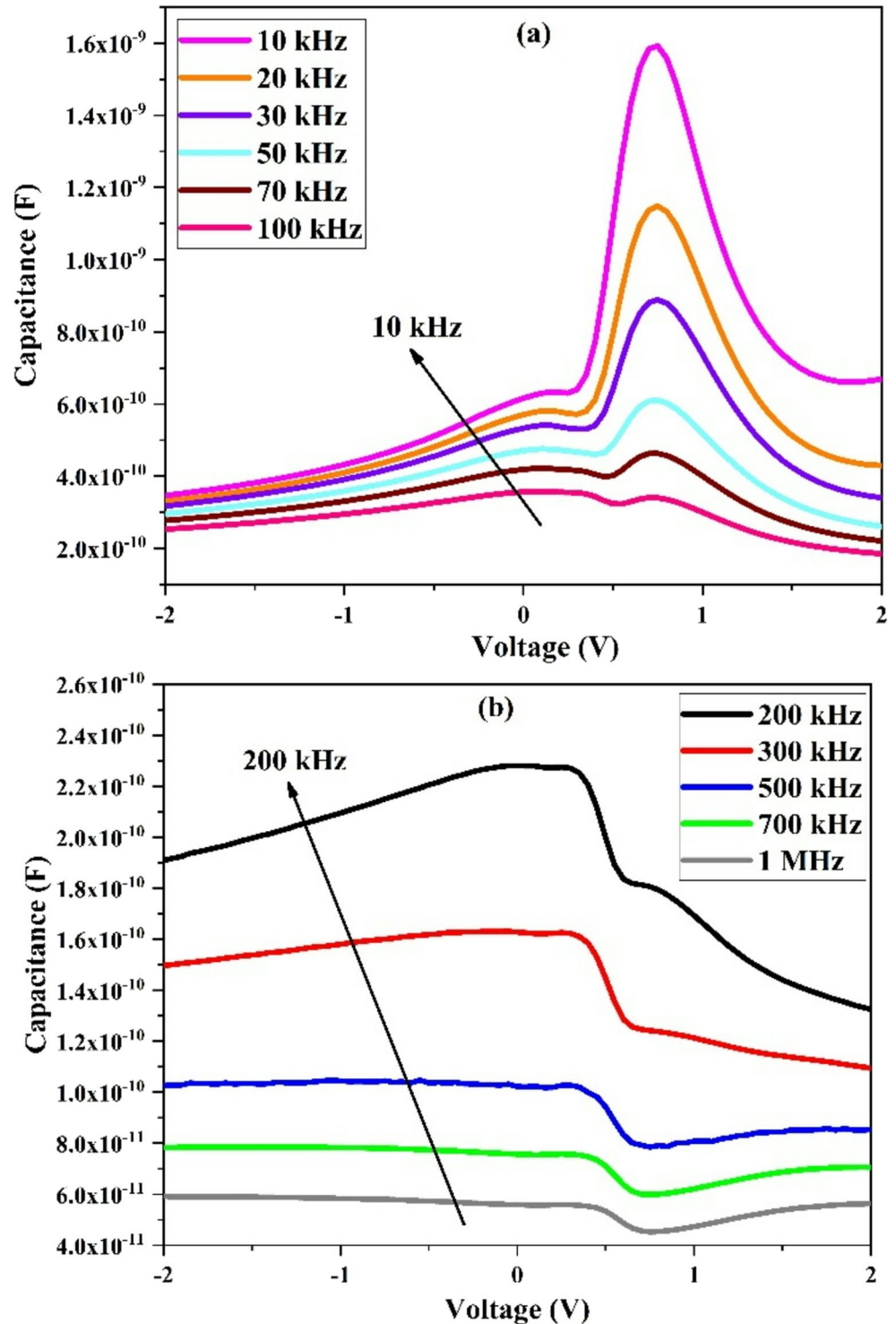
The frequency-dependent capacitance–voltage (C–V) characteristics of the Au/NAMA/n-Si/In MIS structure provide important information about the interface properties, carrier distribution, and polarization mechanisms within the device. Figure 2 (a) and (b) show the measured C–V characteristics obtained at different frequencies in the voltage range of -2 V to $+2$ V.

The frequency-dependent C–V characteristics shown in Figs. 2(a) and (b) exhibit a pronounced dispersion behavior, indicating the strong influence of interface states, interfacial polarization, and charge trapping mechanisms within the device structure [30]. At low frequencies (10–100 kHz), the capacitance values significantly increase and a distinct capacitance peak appears under forward bias conditions. This behavior can be attributed to the ability of interface traps and localized charges to follow the applied AC signal, resulting in additional charge storage at the interface region. As the frequency increases toward the MHz regime, the interface states become unable to respond to the alternating electric field, leading to a substantial reduction in capacitance and suppression of the peak structure.

Moreover, the obtained C–V profiles demonstrate a typical MIS-type (Metal–Insulator–Semiconductor) behavior characterized by accumulation, depletion, and inversion regions. Under negative bias conditions, the device tends to operate in the accumulation regime, whereas increasing positive bias drives the structure into depletion and subsequently toward inversion-like behavior. The transition between these regions becomes more evident at low frequencies due to the enhanced contribution of interface-related charges. The strong frequency dependence and the evolution of the depletion/inversion features confirm the presence of an interfacial insulating layer and electrically active interface states affecting the dielectric response of the device.

In addition to interfacial polarization effects, the observed frequency dispersion behavior may also be influenced by the series resistance and interface-related impedance components of the MIS structure. At low frequencies, interface traps and localized charge carriers can effectively follow the applied AC signal, contributing to higher capacitance values and enhanced dielectric response. However, as the frequency increases, the response of these interface states becomes limited, while the influence of series resistance and bulk impedance effects becomes more dominant. Consequently, the capacitance decreases and the frequency dispersion behavior becomes less pronounced at higher frequencies.

Fig. 2 Frequency-dependent C–V characteristics of the Au/NAMA/n-Si/In MIS structure measured in the voltage range of -2 V to $+2$ V (a) 10–100 kHz and (b) 200 kHz–1 MHz



The observed low-frequency capacitance enhancement can also be associated with charge accumulation at the interfacial layer and interface-related charge effects originating from the NAMA interlayer and native oxide regions. Such behavior is commonly observed in MIS-type structures containing organic interlayers.

For the n-type Si substrate, the applied bias voltage strongly influences the carrier distribution at the semiconductor interface. Under positive bias conditions, electron accumulation occurs near the semiconductor surface, leading to higher capacitance values. As the bias shifts toward negative voltages, the depletion region width increases and

the capacitance decreases. At sufficiently negative voltages, an inversion tendency associated with minority carrier contribution may occur. However, at high frequencies, minority carriers cannot respond effectively to the AC signal, limiting the inversion response.

Although a dedicated forward–reverse C – V hysteresis analysis was not performed in the present study, the frequency-dependent capacitance behavior suggests the presence of charge trapping/de-trapping processes and interface-related trap states within the structure. These trapping mechanisms may contribute to hysteresis behavior in organic interlayer-based MIS structures.

3.2 Dielectric properties

The capacitance–voltage (C – V) and conductance–voltage (G – V) characteristics of the Au/NAMA/n-Si/In MIS-type diode structure were investigated at room temperature over a frequency range of 10 kHz–1 MHz and a voltage range of ± 2 V. Capacitance (C) and conductance (G) measurements as a function of frequency provide important information about interface states and polarization mechanisms in MIS structures. For this reason, this method is widely used in the characterisation of Schottky diodes [15]. The complex dielectric permittivity is used to determine the electrical and dielectric properties of materials. The real (ϵ') and imaginary (ϵ'') components of the complex dielectric permittivity are expressed as $\epsilon^* = \epsilon' - i\epsilon''$, representing ϵ' as the charging current and ϵ'' as the loss current, respectively [31, 32]. The values of ϵ' and ϵ'' are calculated using Eqs. (1) and (2), respectively.

$$\epsilon' = \frac{C_m}{C_0} \quad (1)$$

$$\epsilon'' = \frac{G_m}{\omega C_0} \quad (2)$$

In Eqs. 1–2, G_m represents the measured conductance, C_m represents the measured capacitance, C_0 represents the capacitance of the empty structure, and ω represents the angular frequency. The ϵ' graph of the Au/NAMA/n-Si/In MIS type diode structure as a function of voltage and frequency is shown in Fig. 3a and b.

Figure 3(a) shows the variation of ϵ' as a function of applied voltage at different frequencies. The ϵ' values increase from the negative voltage region to the positive region across all frequency values, reaching a maximum value of around 0.6–0.8 V, whilst they tend to decrease at higher positive voltages. It is thought that this phenomenon is related to the accumulation of charge carriers and the increased contribution of interface states near the depletion

region. At lower frequencies, it is observed that ϵ' values are higher because the interface states are able to track the applied AC signal. An increase in frequency reduces the contribution of polarisation mechanisms, leading to a decrease in the ϵ' value.

Figure 3(b) shows the frequency-dependent variation of the ϵ' value at different voltages. The ϵ' value decreases as the frequency increases across all voltage levels. This behaviour is attributed to the inability of interfacial states and dipolar polarisation mechanisms to respond to high-frequency AC signals. At low frequencies, the higher ϵ' values indicate that interfacial polarisation effects are dominant [33].

Furthermore, the effect of the applied voltage on ϵ' becomes less pronounced at higher frequencies. It is common to find in the literature that ϵ' values change inversely with frequency [34, 35]. In this context, Figs. 3(a) and (b) are quite consistent with the literature data.

Figure 4(a) shows the variation of ϵ'' value as a function of the applied voltage at different frequencies. In the negative voltage region, ϵ'' values are low and close to one another for all frequencies. As the voltage approaches 0 V and shifts towards the positive region, a marked increase in ϵ'' values is observed at low frequencies. This increase indicates a rise in dielectric losses caused by interfacial polarisation and charge trapping at the interface. At high frequencies, ϵ'' values remain low across all voltage ranges. This can be explained by a reduction in energy losses due to charge carriers being unable to respond to high-frequency AC signals.

Figure 4(b) shows the change in ϵ'' value against the frequency applied at different voltages. At all voltage levels, ϵ'' decreases with increasing frequency. The high ϵ'' values at low frequencies indicate significant dielectric losses associated with interfacial polarization and space-charge effects. As the frequency increases, the influence of these polarization mechanisms diminishes, leading to a decrease in ϵ'' values. Furthermore, the effect of different applied voltages on ϵ'' becomes less significant in the high-frequency region. This behaviour can be attributed to the decrease in interfacial polarization contribution with increasing frequency [36]. This behavior can be explained by the Maxwell–Wagner-type interfacial polarization commonly observed in MIS structures. At low frequencies, the dielectric permittivity increases due to the accumulation of charge carriers at the interfacial regions, resulting in higher dielectric loss values. This behavior arises from the ability of interface states and trap levels to follow the applied AC signal. As the frequency increases, the carriers cannot respond sufficiently fast to the rapidly changing electric field, leading to a reduction in interfacial polarization. In the high-frequency region, dipolar polarization and bulk effects become dominant.

Fig. 3 The voltage (a) and frequency (b) dependences of ϵ' for Au/NAMA/n-Si/In structure

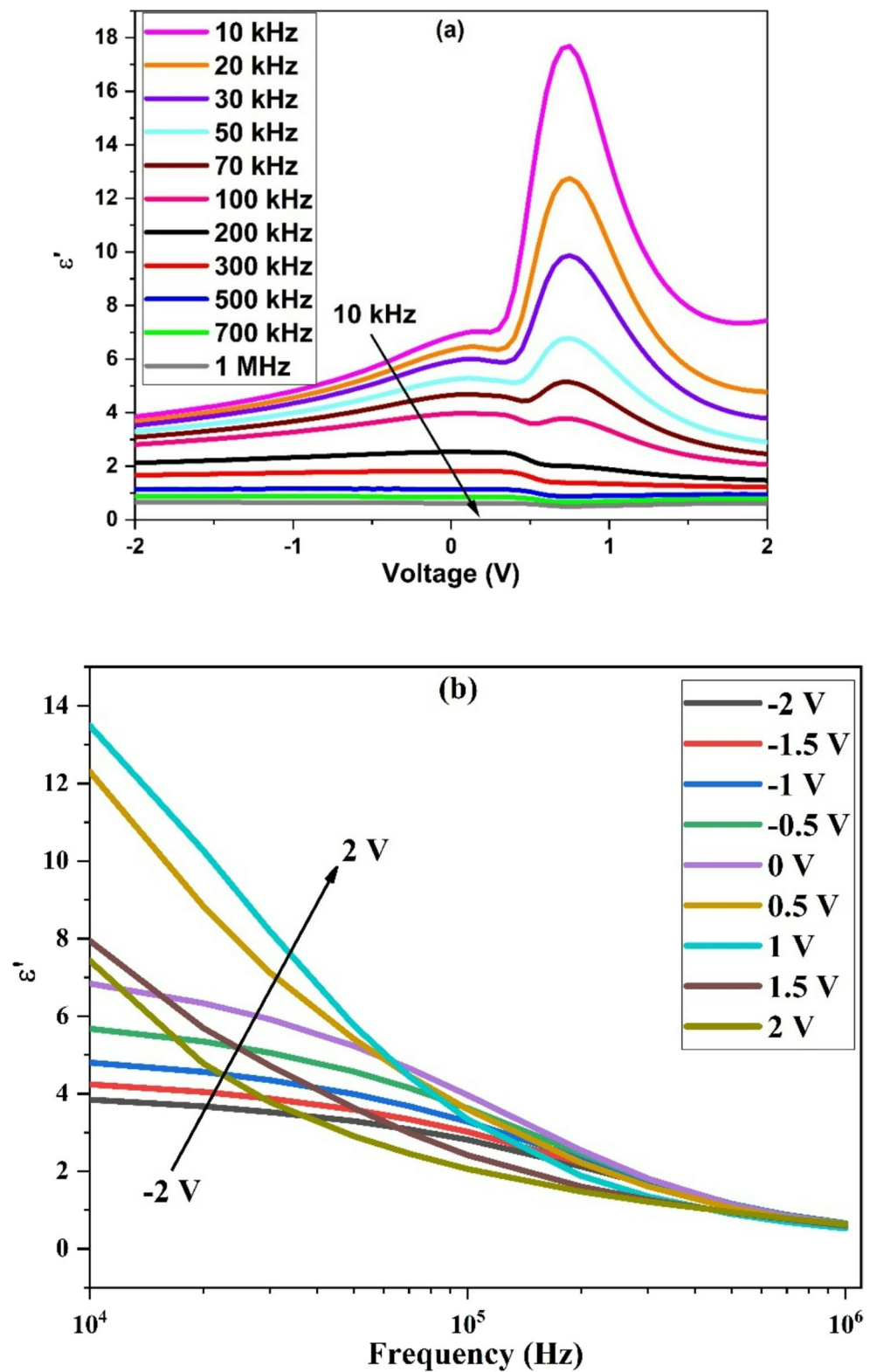
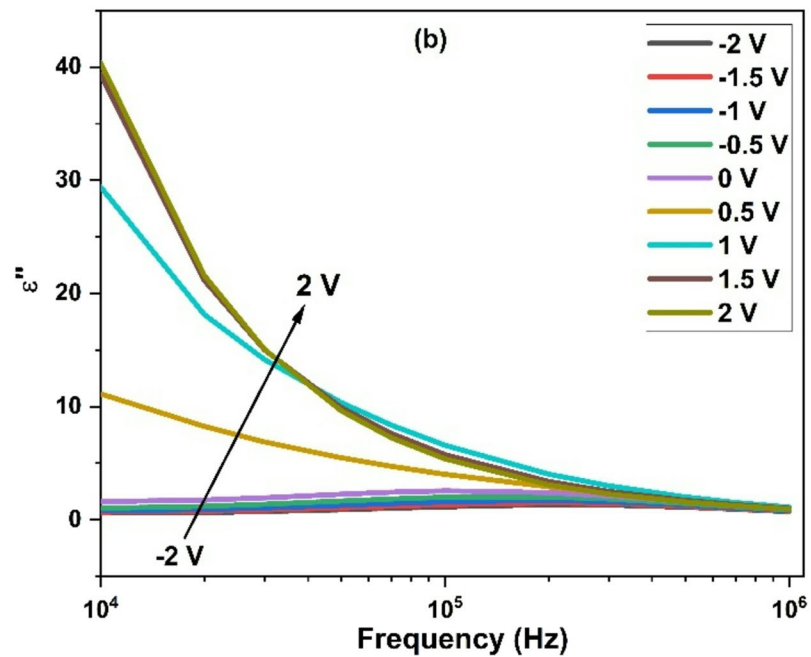
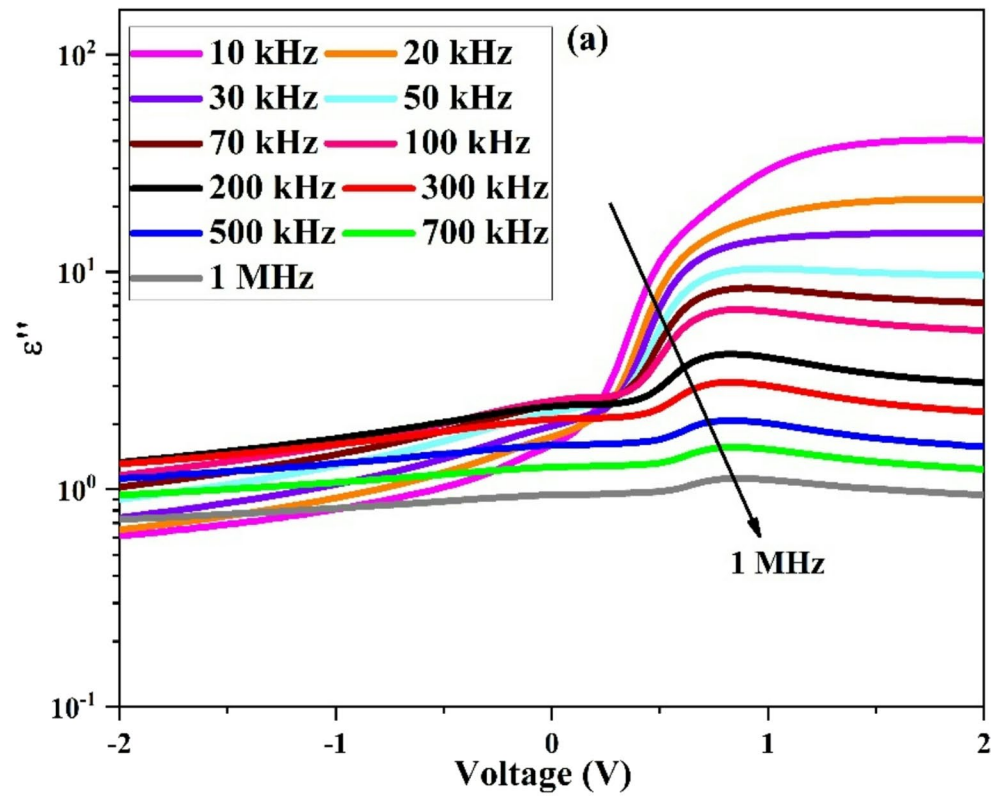


Fig. 4 The voltage (a) and frequency (b) dependences of ε'' for Au/NAMA/n-Si/In structure



The polarisation resulting from the electric field applied to the dielectric material causes intrinsic or extrinsic dielectric losses in the material if it cannot follow the electric field. Dielectric losses are generally expressed as the ratio between ϵ' and ϵ'' , and this ratio is known as the loss tangent ($\tan\delta$) [34]. The loss tangent expression is calculated using Eq. 3.

$$\tan\delta = \frac{\epsilon''}{\epsilon'} \quad (3)$$

Loss tangent ($\tan\delta$) values of samples were determined using Eq. 3. Figure 5(a) shows the change in $\tan\delta$ depending on the applied voltage at different frequencies. In the negative voltage region, $\tan\delta$ values are low and close to each other for all frequencies. As the voltage approaches 0 V, an increase in $\tan\delta$ is observed. In the positive voltage region, particularly in the range of 0.6–1 V, a noticeable increase in $\tan\delta$ occurs. This behaviour can be associated with enhanced dielectric losses due to interfacial polarization and charge carrier motion. As the frequency increases, $\tan\delta$ values decrease, and the curves exhibit lower amplitudes with more limited variation at higher frequencies. This behaviour can be explained by the reduced contribution of polarization mechanisms at high frequencies.

Figure 5(b) shows the frequency-dependent change in $\tan\delta$ at different voltages. In general, $\tan\delta$ exhibits a decreasing trend with increasing frequency, particularly in the low-frequency region. However, at certain applied voltages, deviations from this behaviour are observed, and $\tan\delta$ may show a slight increase at higher frequencies. The relatively high $\tan\delta$ values at low frequencies indicate significant dielectric losses associated with interfacial polarization and space-charge effects [31]. As the frequency increases, the contribution of these polarization mechanisms decreases, and the influence of applied voltage on $\tan\delta$ becomes less pronounced in the high-frequency region.

3.3 Electric modulus analysis

The electrical module mechanism was examined to better understand the relaxation processes of the samples. The modulus is calculated using Eq. 4, which is dependent on dielectric permittivity.

$$M^* = \frac{1}{\epsilon^*} = \frac{1}{(\epsilon' - j\epsilon'')} = M' + jM'' \quad (4)$$

$$M' = \frac{\epsilon'}{\epsilon'^2 + \epsilon''^2} \quad M'' = \frac{\epsilon''}{\epsilon'^2 + \epsilon''^2}$$

Figure 6(a) shows the variation of the M' value with applied voltage at different frequencies. It is observed that M' values are higher in the negative voltage region and decrease with

increasing voltage for all frequencies. A noticeable drop in M' values is observed around 0.6–0.8 V. This behaviour can be associated with the reduction of interfacial effects and the redistribution of charge carriers within the structure. In addition, the overall magnitude of M' increases with increasing frequency.

Figure 6(b) shows the frequency-dependent change in the M' value at different voltages. M' values increase with increasing frequency for all applied voltages. In the low-frequency region, M' values remain relatively low, while a noticeable increase is observed as the frequency increases. This behaviour indicates that electrode and interfacial effects dominate at low frequencies, whereas bulk effects become more dominant at higher frequencies. In addition, M' values are higher at negative voltages, and this difference becomes more pronounced in the high-frequency region.

Figure 7(a) shows the change in the M'' value depending on the voltage applied at different frequencies. In the negative voltage region, M'' values are low and close to each other across all frequencies. As the voltage increases, a distinct peak in M'' values is observed for all frequencies. Following this peak, a slight decrease occurs, after which M'' values show relatively stable behaviour in the positive voltage region. Furthermore, M'' values increase with increasing frequency.

Figure 7(b) shows the frequency-dependent change in the M'' value at different voltages. M'' values increase with increasing frequency across all voltage levels. In the low-frequency region, M'' values are close to each other, and the effect of voltage is limited. As the frequency increases, a marked rise in M'' values is observed, and the difference between the curves corresponding to different voltage levels becomes more pronounced. The peak observed in the M'' curves indicates the presence of a relaxation process in the system. Such peaks in M'' curves have been attributed to dielectric relaxation mechanisms in the literature [34, 35].

The impedance behaviour of the materials is understood by analysing the Cole-Cole curves plotted using the electrical modulus. The $M''-M'$ (Cole-Cole) curves shown in Fig. 8 demonstrate the electrical modulus behaviour of the Au/NAMA/n-Si/In structure at different frequencies. All curves lie within the positive M' and M'' regions. As the frequency increases, the curves shift upward, and M'' values increase. At low frequencies, the curves are narrower and closer to each other, whereas at higher frequencies, they spread over a wider region. In Fig. 8, the curves do not exhibit a perfect semicircular shape. This indicates that the relaxation behaviour deviates from ideal Debye behaviour and suggests the presence of distributed relaxation processes within the structure. To better illustrate the underlying charge transport and polarization mechanisms in the Au/NAMA/n-Si MIS structure, a schematic diagram is presented in Fig. 9.

Fig. 5 Variation of $\tan \delta$ with the applied (a) voltage and (b) frequency

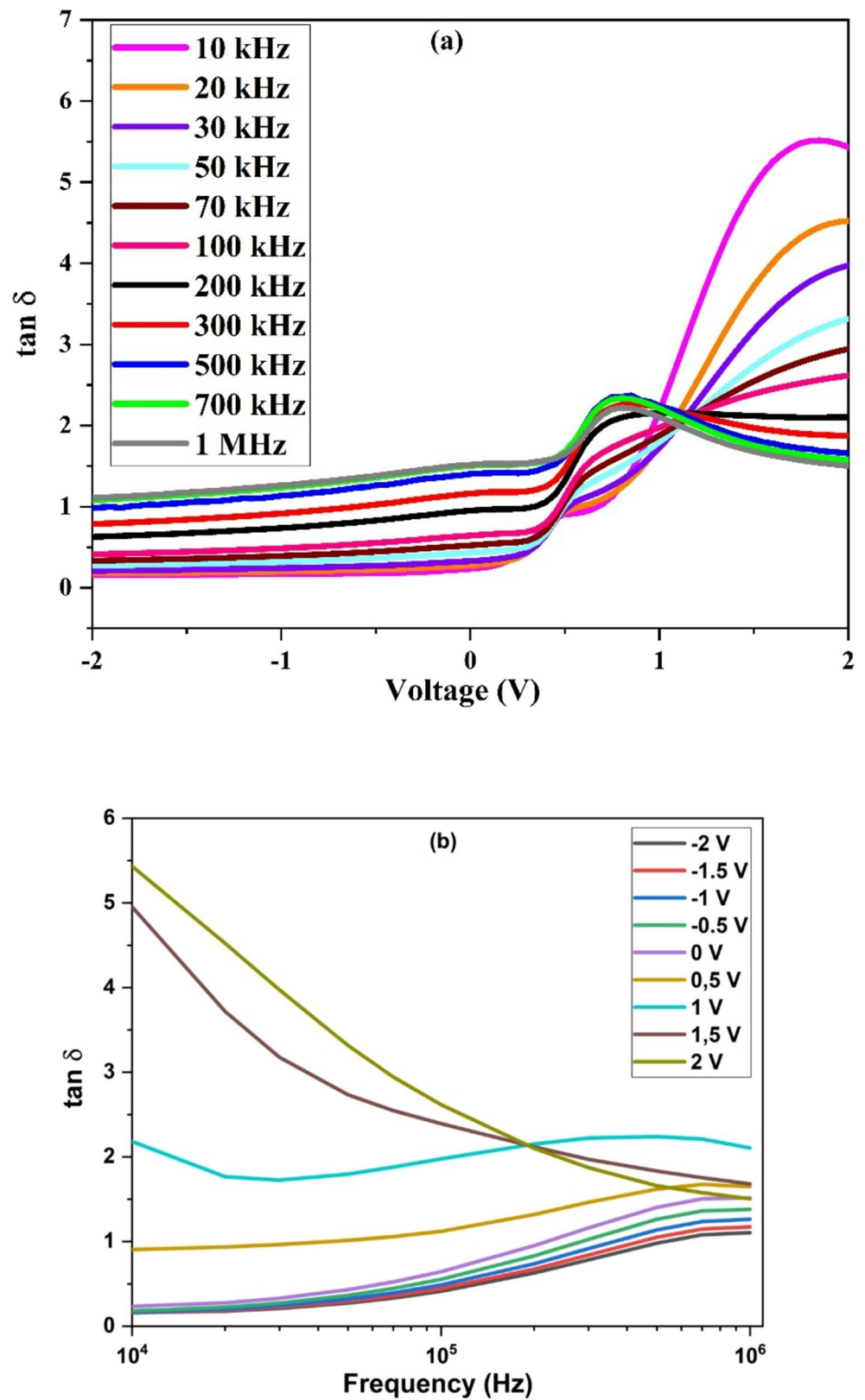


Fig. 6 Variation of M' with the applied (a) voltage and (b) frequency

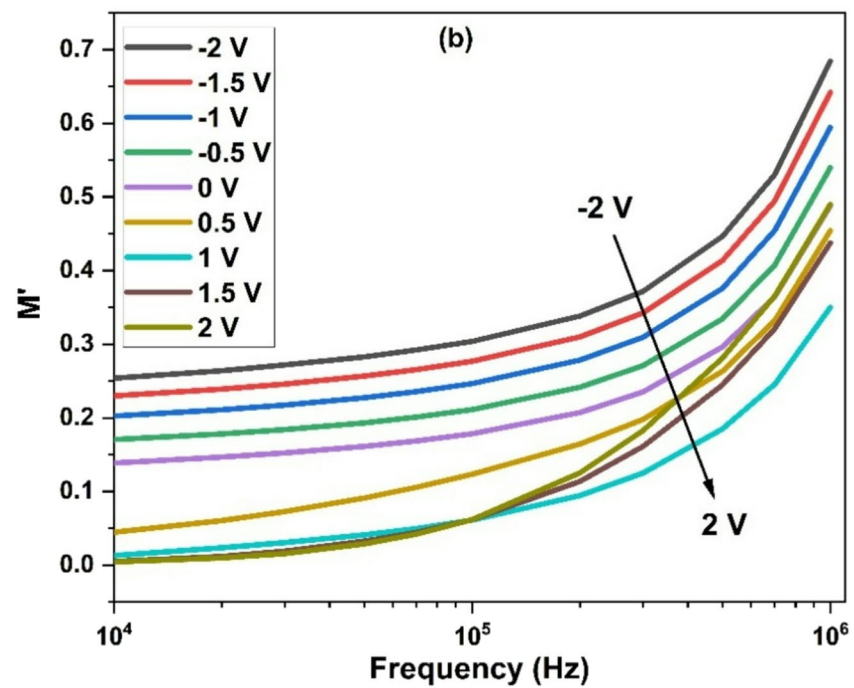
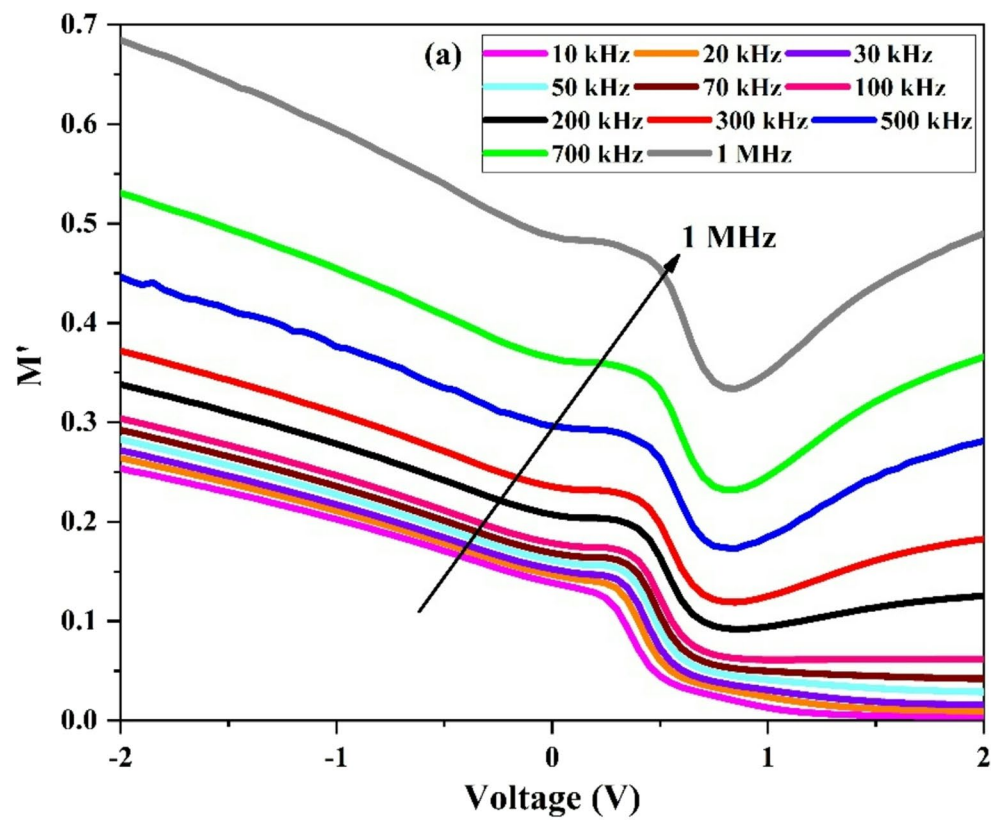
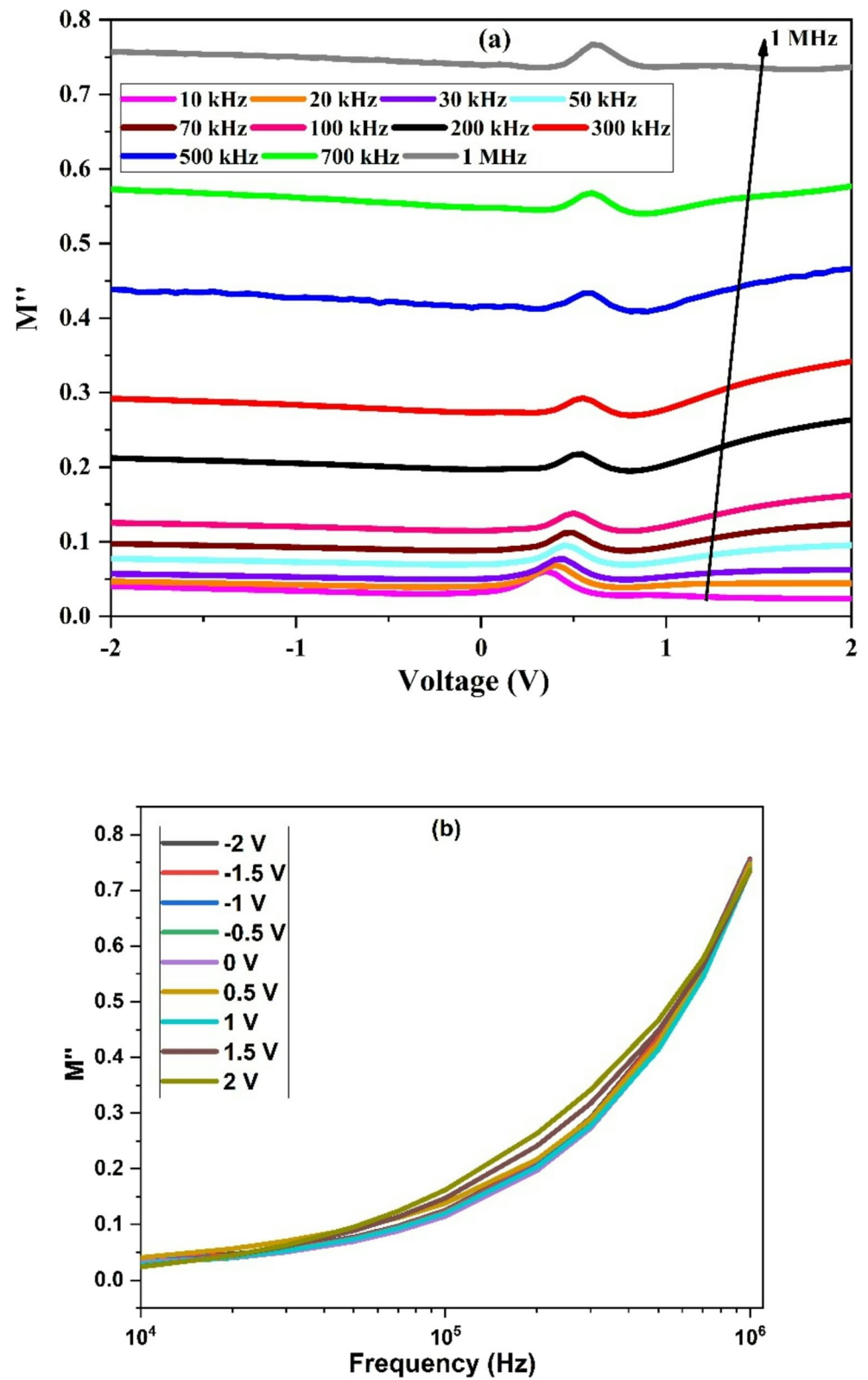


Fig. 7 Variation of M'' with the applied (a) voltage and (b) frequency



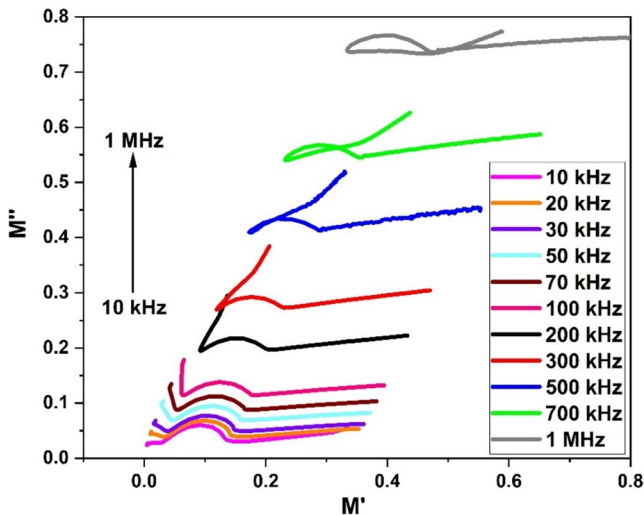


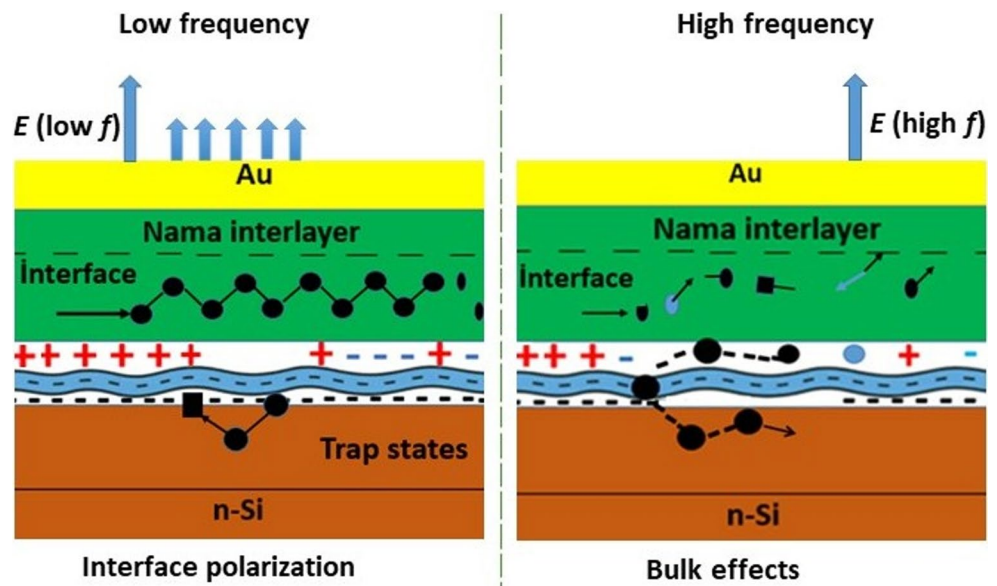
Fig. 8 The Cole–Cole curves for Au/NAMA/n-Si/In structure

As shown in Fig. 9, interfacial polarization and charge accumulation dominate at low frequencies, whereas at higher frequencies, bulk effects become more prominent and the contribution of interface states decreases. In addition, charge transport via hopping between localized trap states contributes to the frequency-dependent electrical behavior.

3.4 AC conductivity

The transmission mechanism for device production and the determination of the carrier type responsible for this mechanism are of critical importance for device performance. Electrical ac conductivity (σ_{ac}) plays a significant role in this selection. Frequency-dependent ac conductivity is determined using Eq. 5.

Fig. 9 Schematic diagram of charge transport and polarization mechanisms



$$\sigma^* = j\epsilon_0\omega\epsilon^* = j\epsilon_0\omega(\epsilon' - j\epsilon'') = \epsilon_0\omega\epsilon'' + j\epsilon_0\omega\epsilon' \quad (5)$$

Parameter σ^* is determined by Eq. 6.

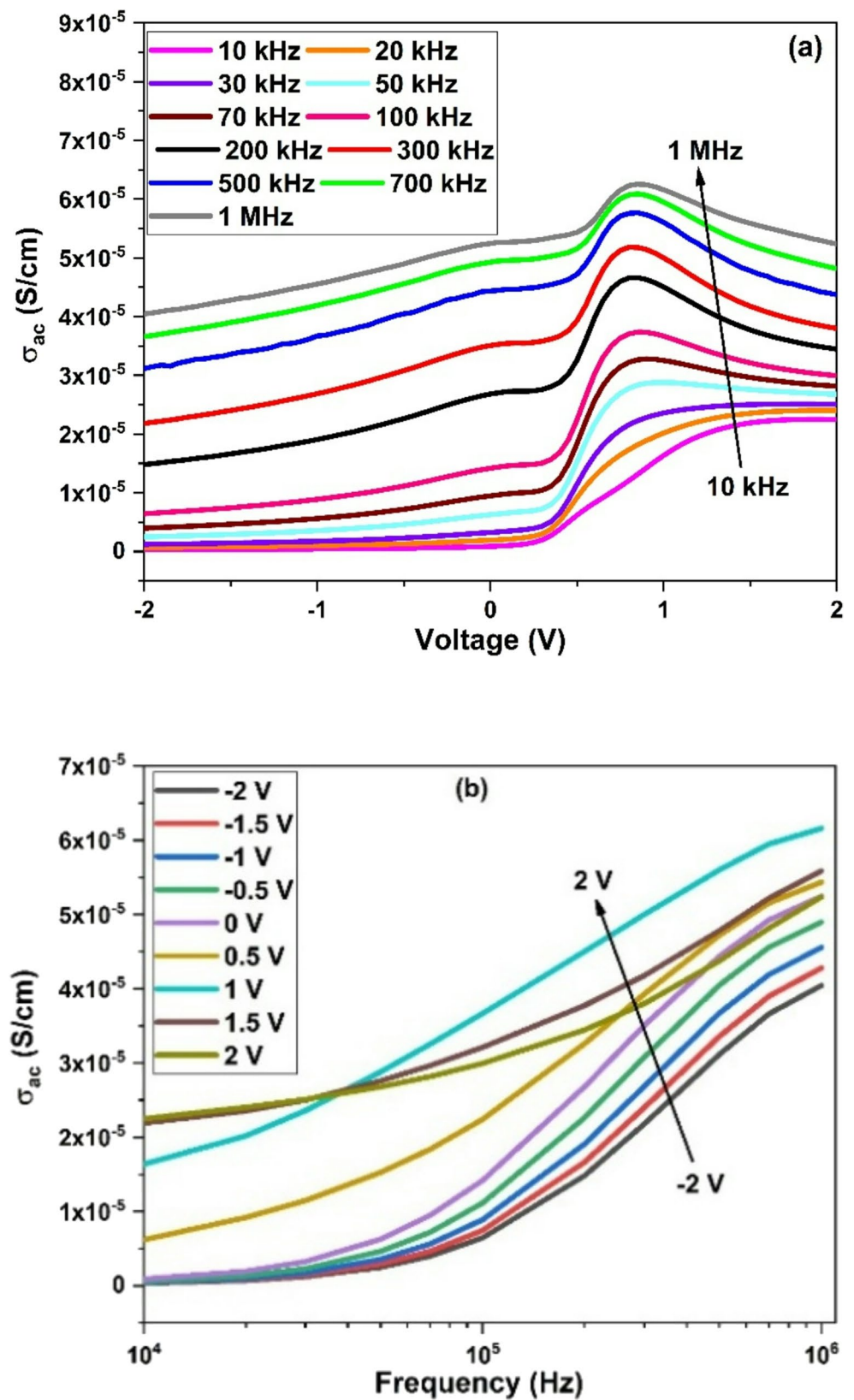
$$\sigma_{ac} = 2\pi f\epsilon_0\epsilon' \tan\delta \quad (6)$$

Figure 10(a) shows the variation of σ_{ac} as a function of applied voltage at different frequencies. In the negative voltage region, σ_{ac} values remain at low levels. As the voltage increases, a noticeable increase in σ_{ac} is observed, particularly in the range of 0.5–1 V. At higher voltage values, a slight decrease in σ_{ac} is observed. In addition, the overall level of σ_{ac} increases with increasing frequency. The maximum σ_{ac} value is obtained as approximately 6×10^{-5} S/cm at 1 MHz (10^6 Hz).

Figure 10(b) shows the variation of σ_{ac} with frequency under different applied voltages. σ_{ac} increases with increasing frequency for all voltage values. In the low-frequency region (around 10^4 Hz), σ_{ac} values remain at low levels, while a regular increase is observed as the frequency increases. In the high-frequency region (around 10^6 Hz), σ_{ac} values vary in the range of approximately 4×10^{-5} – 6×10^{-5} S/cm. Furthermore, σ_{ac} values are higher at positive voltages across the entire frequency range. This behaviour indicates that charge carriers contribute more effectively to conduction as the frequency increases.

The increase in AC conductivity with frequency in semiconductors is a commonly observed phenomenon. The results obtained in this study are consistent with this trend. Indeed, it has been widely reported in the literature that AC conductivity increases with frequency [23]. Furthermore, the increase in AC conductivity with frequency has

Fig. 10 Variation of σ_{ac} with the applied (a) voltage and (b) frequency



been attributed to the hopping of charge carriers between trap levels within the band gap [20]. This suggests that the charge transport mechanism in the Au/NAMA/n-Si MIS structure may be related to trap states.

4 Conclusion

An Au/NAMA/n-Si/In MIS-type diode was successfully fabricated, and its dielectric and electrical behaviors were analyzed using frequency- and voltage-dependent C–V and G–V measurements. The experimental results indicate that the electrical response of the structure is strongly governed by interface-related polarization mechanisms and charge transport processes. The decrease in dielectric permittivity with increasing frequency demonstrates the limitation of interface states and dipolar polarization at high frequencies, whereas the enhanced dielectric loss observed at low frequencies confirms the contribution of space-charge and interfacial polarization effects. The electric modulus formalism revealed relaxation behavior deviating from ideal Debye characteristics, indicating the existence of distributed relaxation processes within the structure. Furthermore, the increase in AC conductivity with frequency suggests that hopping-based charge transport mechanisms may play an important role in the conduction process. The Cole–Cole analysis also supports the presence of non-ideal dielectric relaxation behavior associated with the interfacial layer. Overall, the obtained findings demonstrate that the NAMA interlayer effectively alters the interfacial electrical properties and dielectric response of the MIS structure. These results contribute to a better understanding of charge transport and polarization phenomena in organic interlayer-based semiconductor devices and indicate the potential usability of such structures in tunable dielectric and high-frequency electronic applications.

Acknowledgements The author would like to thank Dr. Serkan Sayın from Giresun University for the help in the synthesis of NAMA compound.

Author contributions Naki KAYA: Conceptualization, methodology, investigation, formal analysis, data curation, writing – original draft, writing – review & editing.

Funding Open access funding provided by the Scientific and Technological Research Council of Türkiye (TÜBİTAK).

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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